

YOANI BINWANA

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Education

University of North Carolina at Charlotte

Bachelor of Science in Data Science

Aug. 2024 - May 2028

Charlotte, Nc

Experience

Tech Fellow

August 2025 – Present

ColorStack

UNC Charlotte

- Led weekly peer coding sessions for 20+ fellows, facilitating real-time debugging and complexity analysis, contributing to multiple internship and research offer outcomes across the cohort.
- Developed performance tracking dashboards to monitor coding frequency and interview progress, increasing consistent weekly LeetCode engagement by 50 percent.
- Designed and implemented a centralized technical prep framework (LeetCode tracker, resume scoring rubric, STAR behavioral template), reducing preparation time by 35 percent and increasing weekly coding consistency.
- Identified low technical interview confidence within cohort and architected a structured mock interview pipeline (DSA rounds + behavioral debriefs), increasing participant pass rates in practice interviews by 40 percent.

Communications Chair

January 2026 – Present

AfroBits

UNC Charlotte

- Led external communications and outreach, initiating partnerships with companies and organizations.
- Identified and developed strategic partnership opportunities to expand organizational reach.
- Built and maintained professional relationships with corporate partners through consistent email communication.
- Collaborated with board members to develop communication strategies that increased student engagement.

Projects

AI Voice Agent Backend | AWS (Lambda, API Gateway, DynamoDB), Python

December 2025 – Present

- Architected and deployed a serverless AI Voice Agent backend using AWS Lambda and API Gateway, reducing infrastructure costs by 60 percent, supporting 10+ concurrent sessions, and maintaining ≤ 300 ms response latency.
- Integrated AWS Lambda-based speech-to-text and text-to-speech pipelines with LLM processing, optimizing asynchronous execution flow to reduce cold start impact and improve processing efficiency by 30 percent.
- Containerized backend services using Docker and configured CI/CD-style deployment to AWS, reducing manual deployment time by 60 percent.
- Implemented structured logging and error monitoring to diagnose latency bottlenecks, improving reliability and reducing dropped sessions by 35 percent.

AI Stock Analyzer | Python, LangChain, REST APIs, Yahoo Finance

May 2025 – Present

- Designed and developed an AI-powered stock analysis engine from scratch, integrating real-time financial APIs with structured LLM-driven analysis to automate equity evaluation.
- Engineered a modular data-processing pipeline using Pandas to compute key financial metrics (P/E, Beta, volatility), reducing multi-stock comparison runtime by 40 percent.
- Implemented dynamic multi-ticker CLI functionality enabling comparative analysis of 5+ equities per execution, improving analytical flexibility and user efficiency.
- Architected structured LLM prompt templates that transformed raw financial metrics into context-aware investment summaries, increasing output consistency and analytical depth.

Stoplight Controller | Python, State Machines, Hardware Prototyping, Finite

December 2025

- Designed traffic light control system using Python with finite state machine (Moore model) architecture to manage bi-directional traffic flow.
- Designed collision-free state machine with 4-state logic (Red/Yellow/Green per direction) ensuring minimum 5-second yellow phase during transitions.
- Validated hardware/software prototype on breadboard, achieving correct state transitions and timing accuracy.

Technical Skills

Languages: Python, Java, C++, HTML/CSS, SQL

Developer Tools: Git, VS Code, PyCharm, Linux, Amazon Web Services (AWS: Lambda, API Gateway, DynamoDB)

Technologies/Frameworks: Linux, GitHub